Module 5

Pregnancy and birth in mammals

Fertilisation

- Fusion of two haploid gametes, forming a single diploid zygote cell, containing the genetic material of both gametes.
- Gametes contain single set of chromosomes (23), and fuse to form a zygote with 46 chromosomes.

Steps in fertilisation

1. Sperm swims using their flagella, through the cervix and into the uterus, then up a fallopian tube where it reaches the egg.
2. Sperm uses enzyme to dissolve the protective layer of the egg.
3. Molecules on the sperm surface bind to cell membrane of the egg to endure its sperm from the same species.
4. Membranes of the sperm and the egg fuse. Changes occur at the surface of the egg to prevent entry of other sperm.
5. Haploid sperm nucleus enters egg cytoplasm, fusing with the haploid egg nucleus and resulting in a diploid zygote.

After fertilisation

- Diploid zygote travels down fallopian tube, until reaches the uterus. During this time, embryonic development occurs. Before implantation, zygote has developed into a blastocyst. Only then is it ready for implantation on the endometrial lining of the uterus.

Hormonal control during menstrual cycle

Two key hormones which control and coordinate menstrual cycle

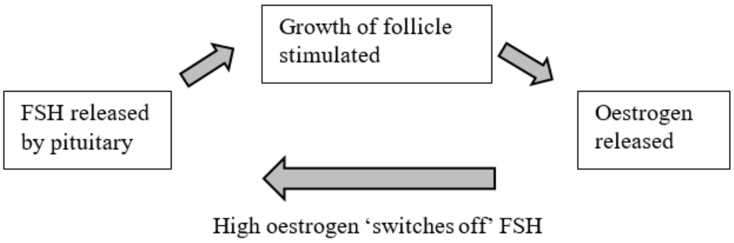
- **Pituitary hormones**
    - Follicle stimulating + luteinizing hormone
    - Released from anterior pituitary gland
    - Act on the ovaries to develop follicles.
- **Ovarian hormones**
    - Estrogen + progesterone
    - Released from ovaries
    - Act on the uterus, prepares for pregnancy

Endocrine Gland	Hormone	Function
Anterior Pituitary 	FSH	• Stimulates follicular growth in ovaries • Stimulates estrogen secretion (from developing follicles)
	LH	• Surge causes ovulation • Results in the formation of a corpus luteum
Ovaries 	Estrogen	• Thickens uterine lining (endometrium) • Inhibits FSH and LH for most of cycle • Stimulates FSH and LH release pre-ovulation
	Progesterone	• Thickens uterine lining (endometrium) • Inhibits FSH and LH

Menstrual cycle steps

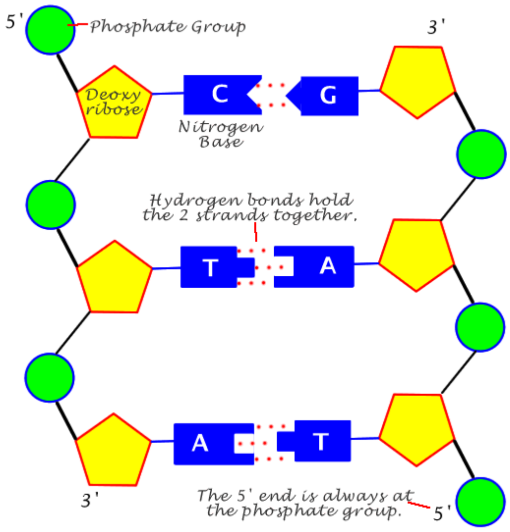
1) FSH stimulates follicular growth.
2) Dominant follicle forms and starts to secrete estrogen.
3) higher estrogen → negative feedback on anterior pituitary → Decreased FSH
4) Day 14: LH surge: → dominant follicle ruptures and egg released (ovulation).
5) Ruptured follicle becomes degenerating corpus luteam which secretes high levels of progesterone
6) Oestrogen and progesterone act on the uterus to thicken the endometrial lining in preparation for pregnancy + inhibit the secretion of FSH and LH preventing development of any other follicles.

Negative feedback in menstruation.



Cell replication

Composition of DNA.



Advantage of complementary base pairs

Double stranded DNA will always have a backup copy of all genetic information so our body can repair the damage to the DNA. E.g. UV light damages DNA, body can use nucleotide sequence of DNA in one strand to fix the other.

Watson and Crick Model

- Double stranded, anti-parallel helix structure
- Sugar-phosphate backbone makes up the outside of the helix
- Nitrogenous bases are inside, and form hydrogen bond pairs that hold the strands together.

How DNA is packaged

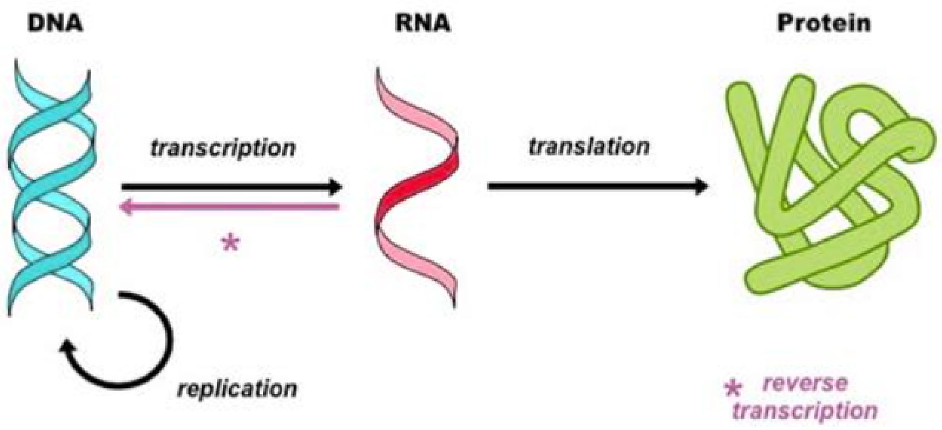
- Double stranded A-T, G-C pairing.
- DNA is wrapped around histones to form nucleosomes.
- Nucleosomes fold to form fibres
- Fibres form loops
- Looped fibre is tightly coiled, making up a chromosome.

DNA replication

Semi conservative: each DNA strand serves as a template for the synthesis of a new strand.

1) DNA helicase unwinds DNA and breaks hydrogen bonds.
2) DNA primase lays down RNA primer that provides a free -OH at the 3' end for the DNA polymerase to act upon.
3) DNA polymerase adds on nucleotides to a free -OH at the 3' end of the DNA strand, forming complementary strand.
4) Different form of DNA polymerase removes RNA primer and replaces it with DNA.
5) DNA ligase fills in the nicks.

Polypeptide synthesis

DNA	RNA	PROTEIN
		
Located inside the nucleus	Formed in the cytoplasm but can move in and out of the nucleus	Located only on the nucleus
2 strands of genetic mater	A copy of one section of DNA only	Functional molecules in the template of the gene of the chromosome.

Transcription

Synthesis of RNA from a DNA template → code in the DNA is converted into a complementary RNA code.

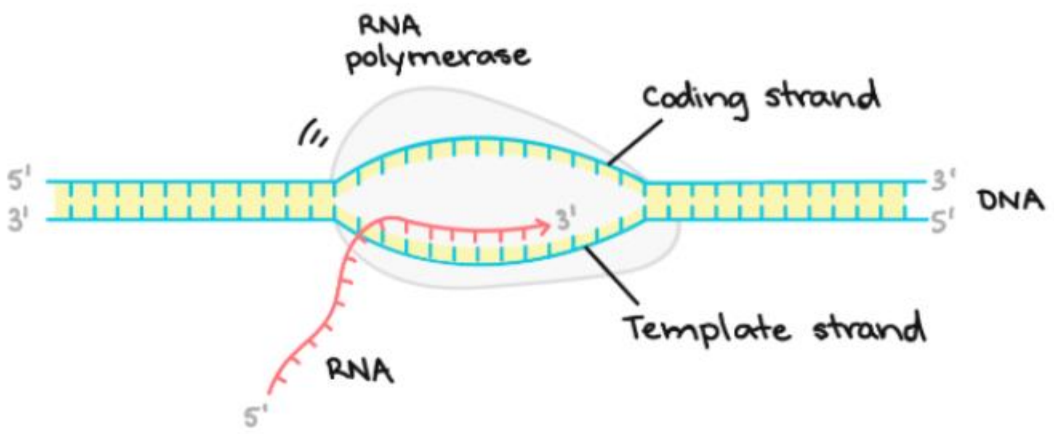
Translation

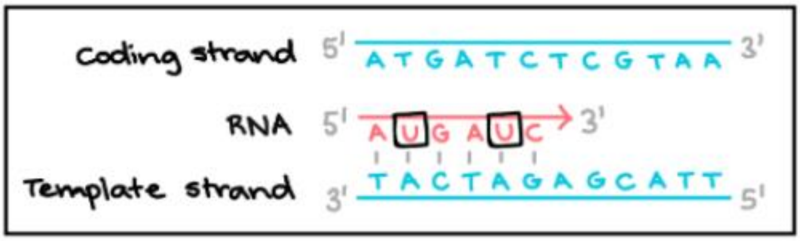
Synthesis of a protein from an RNA template → where the code in RNA is converted into amino acid sequence in a protein.

Steps to transcription

Initiation

- RNA polymerase binds to promoter region and separates DNA strands.





Elongation

- RNA polymerase reads template one base at aa time and builds RNA molecule- "elongating RNA strand".
- RNA strand contains uracil instead of thymine.

Termination

- Sequences called terminators signal completion of RNA transcript
- RNA strand is released from RNA polymerase.
- mRNA that was transcribed leaves nucleus and travels through cytoplasm, where it binds with a ribosome.

Before translation, transcribed RNA must be processed. This means removing and adding specific sections of RNA. E.g. Splicing

Splicing: non translated regions of the gene called introns are spliced out of the mRNA and translated regions called exons are stuck back together.

Steps to translation

1) Ribosome moves along the mRNA molecule and attach tRNA molecules to them by pairing the bases of the tRNA's anticodons with their complimentary triplets of bases in the mRNA.
2) Amino acids from the tail end of the tRNA link to each other to form a polypeptide chain. Each amino acid is then spliced off its tRNA carrier.
3) tRNAs move away from mRNA. Leaving the growing chain of amino acids and move back into the cytoplasm where they can pick up another amino acid and be reused.

Types of RNA that are transcribed

1) Messenger RNA (mRNA): carries complimentary copy of DNA that specifies the amino acid sequence of a polypeptide. Synthesised in the nucleus, from which it travels to the cytoplasm and binds to a ribosome for translation.
2) Ribosomal RNA (rRNA): together with proteins, forms an organelle called a ribosome, which are required in the translation of mRNA.
3) Transfer RNA (tRNA): transferring amino acids from the cytoplasm to the ribosome, where they are linked to form a polypeptide chain based on the sequence of nucleotides in the mRNA.

Translation is halted via a stop codon. No tRNA molecules can recognise a stop codon so translation is terminated.

Structure and function of 3 proteins

Proteins: macromolecules consisting of one or more sequences of amino acids called polypeptide chains.

Types of proteins

Signalling proteins (hormones): proteins released from glands that travel through the circulation to target organs and regulates their behaviour. E.g. insulin and glucagon

Structural proteins: provide a necessary stiffness to fluidlike cells. E.g. collages and elastin are essential components of connective tissues such as cartilage

Protein	Structure	Function
Haemoglobin	made up of four globular polypeptides, shows a quaternary structure. Contains iron ions.	Iron ions have high affinity to oxygen making haemoglobin ideal for transporting oxygen.
Keratin	Fibrous protein that comprises three linear polypeptide chains that wind around each other	provides strength to the structure. Provides a smooth, flexible yet resilient padding that protects the ends of long bones at the joints
enzymes	The molecular structure of the active site of the enzyme either matches (lock-and-key model) or conforms to (induced fit model) the structure of the substrate	Allow the enzyme to bind to the substrate and catalyse a reaction

Mitosis and meiosis

Mitosis

INTERPHASE

- **DNA replication occurs** in the cell so that each daughter cell will have a full set of DNA.
- Other **preparations for mitotic division** e.g. enlarging and duplicating of cell structures.

PROPHASE

- The cell's **nuclear membrane begins to disappear**.
- **Chromatin condenses, forming chromosomes**: two sister chromatids joined at the centromere.

METAPHASE

- The **chromosomes line up along the equator**
- **Spindle fibres attach** to chromosome centromere in preparation to separate.

ANAPHASE

- **Sister chromatids separate at the centromere**, and each individual **chromosome pulled to either side of the cell**.

TELOPHASE

- New **nuclear envelope begins to form** around each set of chromosomes.
- Chromosomes unwind to form chromatin.

CYTOKINESIS

- **Cell divides** to form two genetically identical daughter cells.

Meiosis

PROPHASE 1: CROSSING OVER OCCURS HERE

- **Chromosomes separate to form homologues pairs.**

METAPHASE 1

- The **chromosomes line up along the equator**
- **Spindle fibres attach** to chromosome centromere.

ANAPHASE 1

- **Sister chromatids separate at the centromere**, and each individual **chromosome pulled to either side of the cell**.

TELOPHASE 1

- New **nuclear envelope begins to form** around each set of chromosomes.
- Chromosome number for each cell is halved

CYTOKINESIS 1

- **Cell divides** to form two genetically DIFFERENT daughter cells.

PROPHASE 2

- Spindle fibres reform and attach to centromeres.

METAPHASE 2

- The **chromosomes line up along the equator**
- **Spindle fibres attach** to chromosome centromere.

ANAPHASE 2

- **Chromatids move apart, to opposite sides of the cell. Spindle disintegrates.**

TELOPHASE 2

- Four haploid nuclei begin to form, with single chromatids.

CYTOKINESIS 2

- Four haploid daughter cells have now formed. All of which are genetically unique.

Independent assortment: genes are passed independently of other traits from parents to offspring. E.g. hair colour is passed independently of eye colour.

Random Segregation: copies of a gene separate so that each gamete only receives one allele of that gene.

Complete and Incomplete Dominance

Genes

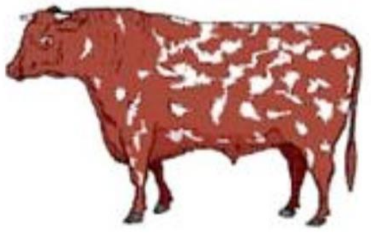
Segment of DNA on a chromosome that specifies a particular characteristic.

Alleles

Alternative forms of the same gene

Codominance

Heterozygous organisms in the first generation show phenotypes of both alleles in the gene concerned. E.g. the roan cow, where both red and white coat colour were expressed. Or blood types. Person with genotype I^AI^B will have blood type AB, as both alleles are expressed presenting co dominance.



Incomplete dominance

Both alleles of a pair are only partially expressed in the heterozygote, e.g. snapdragons have a red and a white allele for flower colour and the red allele is only partially expressed to give the phenotype pink.

DNA sequencing and profiling

There are three techniques involved in sequencing

- Molecular genetic testing: Comparing and contrasting segments of DNA
- Cytogenic testing: identifying abnormalities in the number and structure of chromosomes.
- Biochemical genetic testing: identifying the amount or activity level of certain proteins.

DNA technologies

PCR (polymerase chain reaction)

Technique used to make many copies of a specific gene invitro, to perform tests or research or be analysed in some way.

Could be used to replicate a gene that is related to cancer or be used in forensics to compare crime scene DNA with a suspect.

Steps:

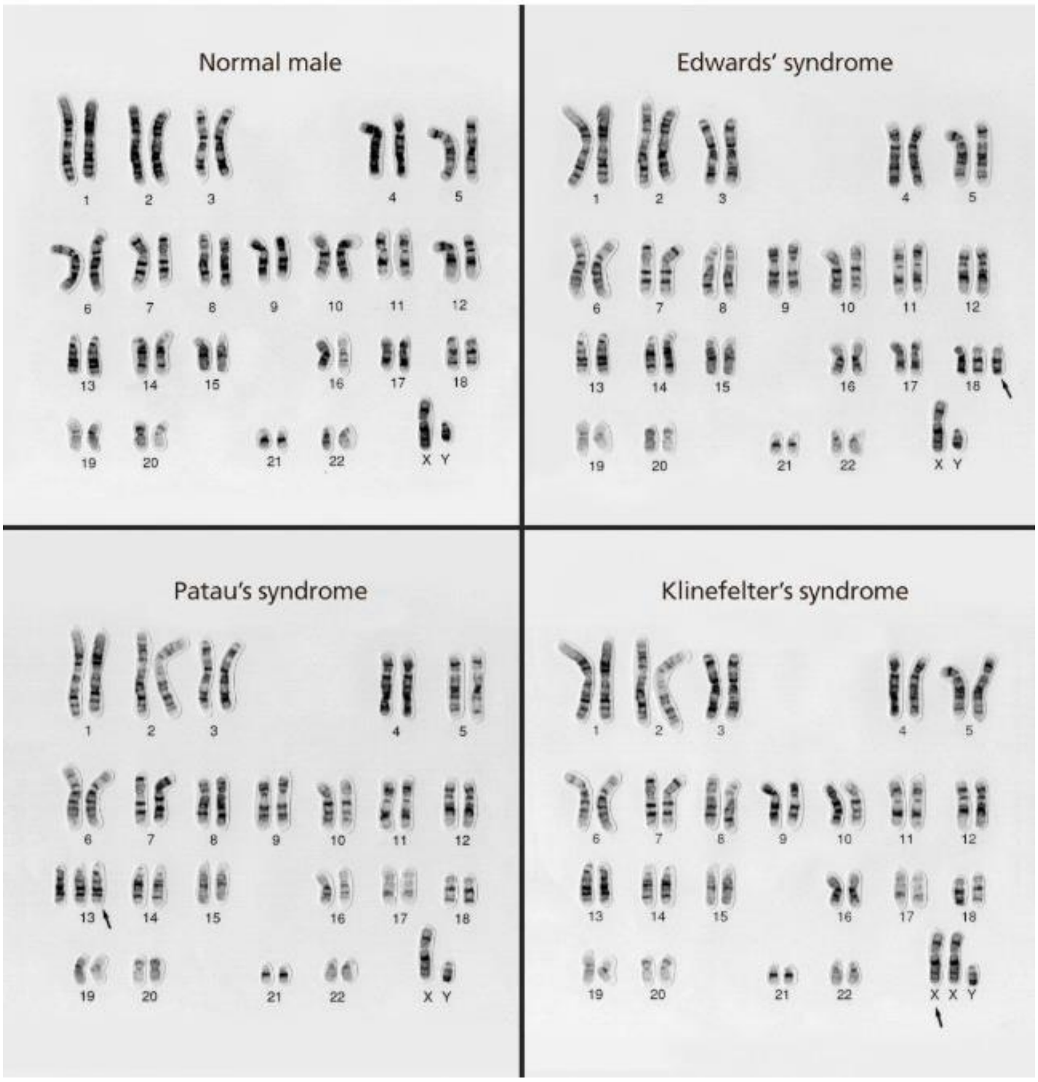
1) DNA strands are heated to denature and thus separate the strands. This provides single stranded DNA for the next step.
2) Mixture is cooled so that primers can bind to their complementary sequence on the single stranded template DNA.
3) Mixture is heated again to the optimal temps of Taq polymerase.

DNA sequencing

- Process of determining sequence of nucleotides in a piece of DNA.
- Similar to PCR, but special nucleotides are added to the mixture. Each is tagged with a fluorescence dye.
- DNA segments are injected into a capillary tube
- Large voltage is applied causing negatively charged DNA segments to move towards positive electrode
- Segments move through a laser beam causing them to glow at a specific colour depending on their base. Order at which bases pass through the beam allows the sequence of the bases to be determined.
- The data is fed to a computer where it can be analysed.

Karyotyping

Involves using a light mocroscope to obtain an individuals set of chromosomes. Useful for detecting large scale changes in the chromosome.



Module 6

Mutations

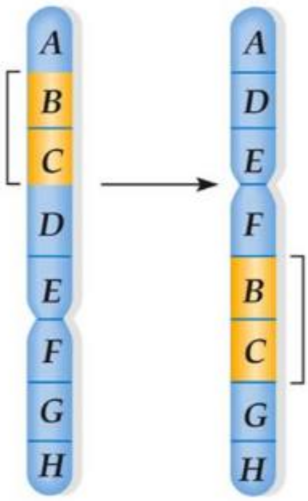
Change in the structure of DNA, can be naturally occurring or caused by mutagens e.g. physical, chemical, biological

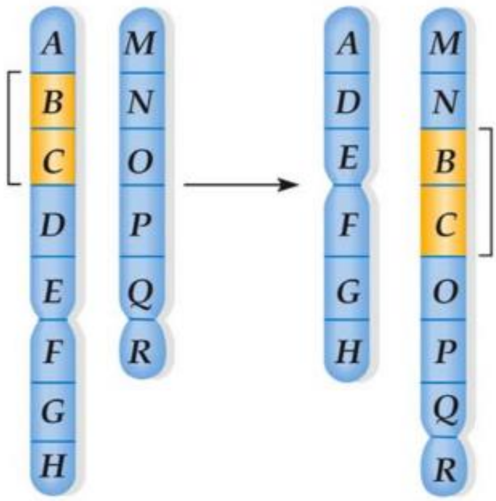
Chromosome mutations

Result from changes in the chromosome number or rearrangement of genes on chromosomes. Typically affects multiple genes. Tend to occur during meiosis in eukaryotic cells. Tends to be fatal, or causes severe retardation.

Five main types of chromosome mutations

- **Duplication**
  Same section of the chromosome is copied or repeated. May occur because a piece of chromosome breaks off and joins onto its homologous chromosome. Often accompanies deletion. E.g. **Charcot Marie tooth disease.**
- **Inversion**
  Occurs when order of the genes are reversed. Generally not harmful , as genes remain accounted for and unchanged.
- **Deletion**
- Part of the chromosome breaks off completely and is lost. E.g. **cri du chat syndrome**
- **Translocation**
  When part of one chromosome breaks off and attaches to another chromosome. Can occur within the same chromosome or from one chromosome to another.

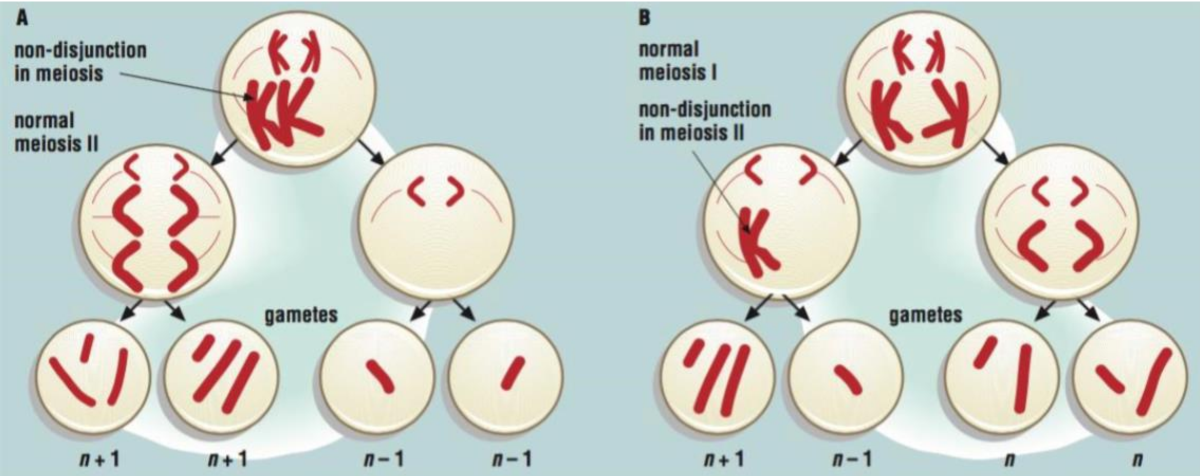




Chromosome abnormalities.

Trisomy

Sometimes, during meiosis, chromosomes do not separate completely. This is non disjunction.



Second pair of homologous chromosomes have failed to separate. Both move to one side of the cell resulting in two gametes having an extra chromosome and two having one fewer.	Non-disjunction in meiosis 2, resulting in one gamete having an extra chromosome, and one having a chromosome less. Other two are normal.

When gamete with extra chromosome is fertilised by a normal zygote, it now has three of a particular chromosome instead of two. The mutation is then inherited by all cells in the embryo as it is copied during mitosis. This is **trisomy.** e.g. Klinefelter's syndrome: when male has two X chromosomes + one Y.

Polyploidy

Occurs when all the chromosomes fail to separate. One gamete has all the chromosomes, while another has none. Produces a gamete with 2n chromosomes. When its fertilised with a normal gamete with n chromosomes, produces a zygote with 3n chromosomes. This is a triploid cell. Much more common in plants than animals, as a polyploidy in animals tends to be fatal.

Point mutations

When one nucleotide in DNA is replaced with another nucleotide.

- **Silent mutations**
  When a nucleotide substitution results in a new codon that codes for the same amino acid. Causes no change in protein produced.
- **Missense mutations**
  Nucleotide substitution results in new amino acid. Produces a different protein and may not function properly. E.g. sickle cell anaemia.
- **Nonsense mutations**
  When nucleotide substitution results in the formation of a stop codon. This signals for translation to stop, terminating protein synthesis.

Frameshift mutations

When a base is added or deleted which alters the entire sequence of DNA after the mutation. This changes all DNA triplets and hence mRNA codons. Sequence of many amino acids is affected.

Occurs in muscular dystrophy.

Mutations in non-coding regions

Non-coding regions include:

- Introns
- Non-coding RNA's
- Promoter regions
- Terminator regions

Intron mutations

Before transcription, introns are spliced out by spliceosomes to create a functional mRNA. After its removed, exons join together to create an mRNA molecule with a continuous coding sequence.

- ➔ If mutations occur at spliceosome binding site, splicing may fail, and intron will be retained in mRNA. This results in introns being read as coding sequence in the production of mRNA's, causing other mutations. This is a pseudo exon.
- ➔ Diseases caused may include cystic fibrosis, muscular dystrophy.

Promoter and terminator mutations

Promoter region is where RNA polymerase binds to initiate transcription. Terminators are sequences that signal to end transcription. Mutations in these regions usually result in ovefr or under expression of a gene.

Population studies

Gene pool

- Collection of all alleles of all genes found in a freely interbreeding population.

Gene flow

- Movement of alleles between populations

Genetic drift

- Changes in allele frequency due to random chance e.g. births or deaths. In small isolated populations, random loss of alleles can have a significant effect, resulting in loss of genetic variation.
- *Founder effect:* genetic drift as a result of several individuals in a population becoming geographically isolated from the rest of the population.
- *Bottleneck effect*: genetic drift due to natural disaster.

Cloning

Whole organism cloning

Involves cloning the entire organism. It is commonly used in research, where genetically identical test subjects are required for reliability. Its also used in agriculture where cloning enables desirable characteristics to be passed onto many offspring in a method faster than traditional breeding.

<u>Two types of whole organism cloning are:</u>

- Artificial embryo twinning
- Somatic cell nuclear transfer (SCNT)

Artificial embryo twinning

- Technique that **imitates the natural process that leads to twins**. Naturally twins form early in development when the embryo splits in two.
- Twinning occurs in the first days after fertilisation, while the embryo is made of a small number of unspecialised cells. Each half of the embryo continues developing on its own, eventually developing into two complete genetically identical individuals.
- Artificial twinning uses the same approach, but its carried out in vitro.

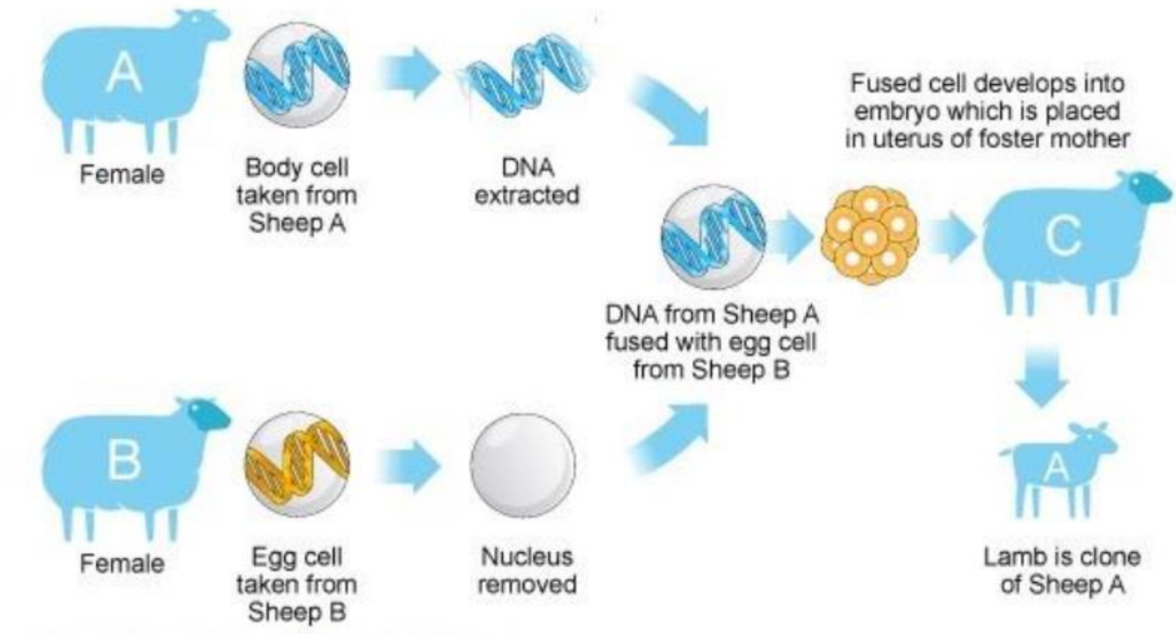
<u>Steps in artificial embryo twinning</u>

1) Very **early embryo is separated into individual cells**, which are **allowed to divide** for a short time in a petri dish.
2) Embryos are then **placed inside a surrogate mother**, where they **finish developing**.
3) Since all embryos came from the same fertilised egg, they are **genetically identical**.

Somatic cell nuclear transfer (SCNT)

Steps:

1) **Nucleus in egg cell is removed** and **replaced with the nucleus of somatic cell** from the organism being cloned.
2) **Egg cell is induced to divide** as though under natural fertilisation. (sperm not required as egg cell already has 2n chromosomes from somatic cell)
3) **After development** of embryo, it **is transferred into the uterus of a surrogate mother**.
4) Resulting individual is a **genetically identical copy of the animal** from which somatic cell was taken.



The cloning process of 'Dolly the sheep'

Therapeutic Cloning

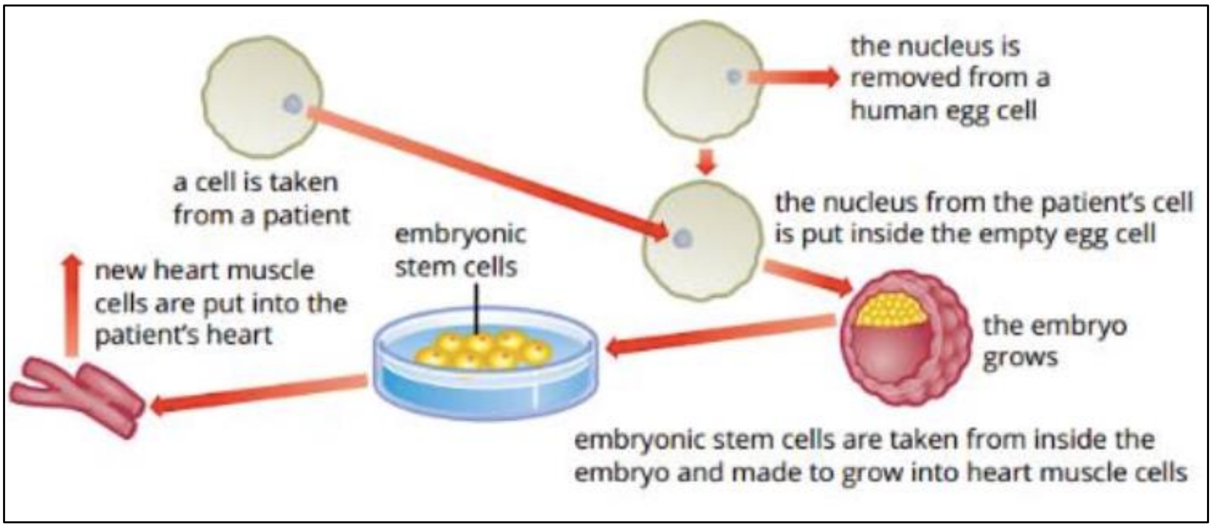
Stem cells

Unspecialised cells with the potential to develop into any form of specialised cell e.g. muscle or nerve cells.

- Therapeutic cloning uses SCNT to clone embryonic stem cells for the research and treatment of disease.
- Much ethical debate about the use of SCNT on human cells for reproductive purposes, thus reproductive cloning of humans is currently banned.
- Therapeutic cloning has the potential to overcome issues regarding immunological rejection of cells if patients own stem cells are used for their treatment.

<u>Steps in therapeutic cloning</u>

1) Nucleus of a skin cell (usually skin cell) inserted into a fertilised human egg whose nucleus has been removed.
2) Egg then begins to divide rapidly and eventually develops into a blastocyst.
3) Stem cells are extracted from the blastocyst and can be made to grow into specific cells e.g. muscle.



Recombinant DNA

Plasmids

Small circular DNA molecules found in bacteria. They are self-replicating, stable and easy to manipulate in a laboratory environment.

Vector

A vehicle/ vessel used to transfer genetic material to a target cell.

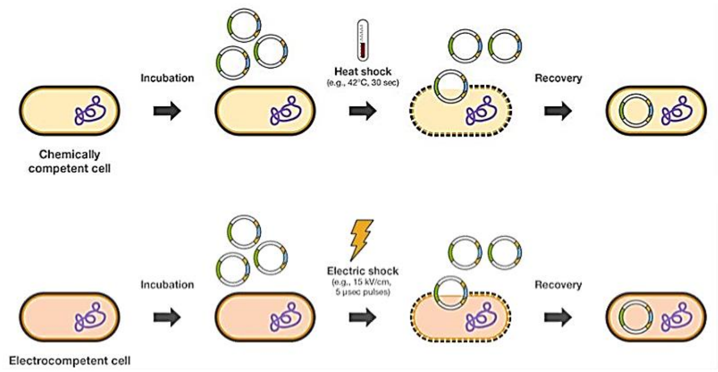
Steps in creating recombinant plasmids

1) Target **DNA is cut** from cell **using restriction enzymes**.
2) Bacterial **plasmid is cut** using the **same restriction enzyme**. Plasmid and DNA now have **exposed DNA bases** that are **complementary to each other**.
3) **Plasmid and target DNA are placed together**. Some plasmids will close back up (non-recombinant plasmids), while others will bind with the target DNA by complementary base pairing.
4) **DNA ligase is added** to re-j**oin the sugar phosphate backbone** of the DNA.

Transforming bacterial cells

Incorporation of the recombinant plasmid into the bacterial cell. Two methods to do so:

1) **Heat shock:** bacteria and plasmids are placed in cold solution. Temp is rapidly heated disrupting the cell membrane of bacteria, allowing the plasmid to enter.
2) **Electroporation:** bacteria and plasmids are subject to electrical current that also disrupts cell membrane, allowing plasmid to enter.



Selection of transformed bacteria

- Recombinant plasmids had a specific antibiotic resistance gene inserted into them, while plasmids that did not take up the target DNA (non-recombinant plasmids) do not.
- Bacteria is incubated on agar plates, containing the antibiotic. Bacteria that have the resistance gene survive and others will die out.

Transgenic organisms

Any organism that has had its **genome altered in any way** is a **genetically modified organism** (GMO). An organism that has had **genes from another species inserted into its genome** is called a **transgenic organism**.

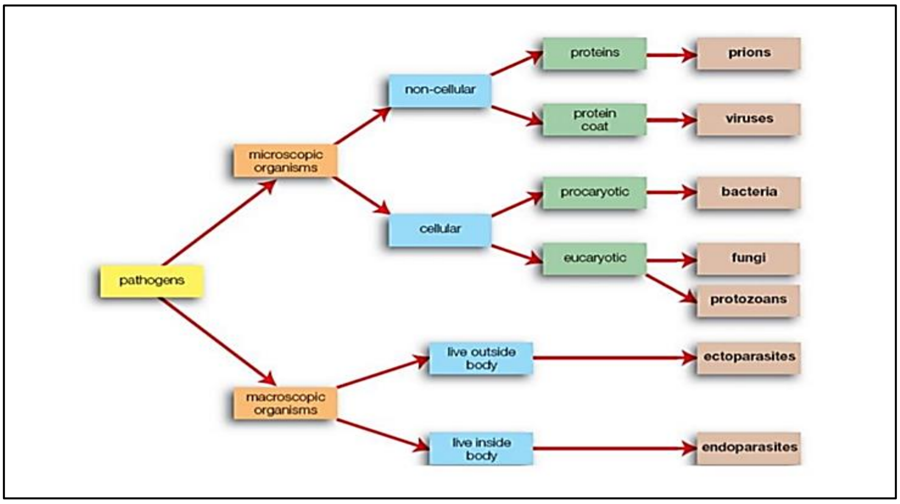
Bt cotton

- **Cotton** is a plant that **attracts many pests**. To protect them, **plants are sprayed with high amounts of insecticide** which may harm beneficial insects, can even impact human health and are quite costly to use.
- **Bt cotton** is a transgenic crop that has been modified to **contain genes** from the soil **bacteria Bacillus Thuringiensis**. Expression of the gene **produces proteins** in the plant that **kill the main caterpillar pest** of cotton.
- This has drastically **reduced insecticide use** and has allowed farmers to **decrease environmental impact** as well as reduce costs.
- Bt cotton has been in use in Australia for over 15 years now, and scientists have found **no adverse effect** on the environment.

Module 7

Info on pathogens

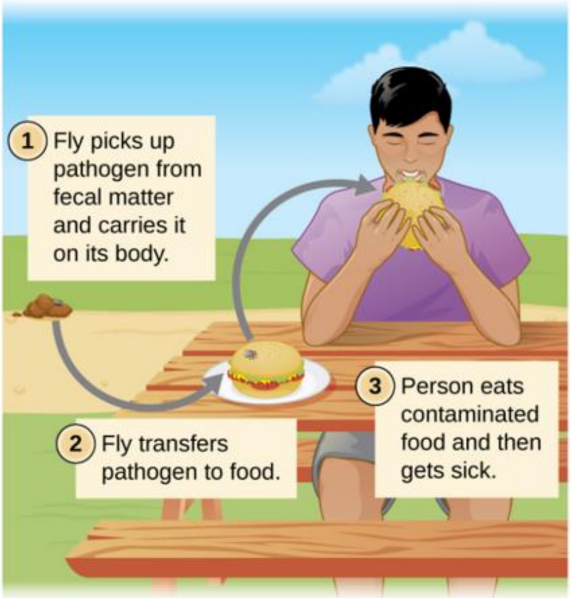
PATHOGEN	CHARACTERISTICS	DISEASE EXAMPLES
Bacteria	• Single celled and no membrane bound organelles. • Prokaryotic. • 0.5 to 100 micrometres.	• Anthrax - infectious disease caused by bacteria bacillus anthracis.
Prion	• Protein. • They are non-cellular. • No genetic material. • A mutation in the gene that codes for normal prion proteins, causes the production of disease causing prions.	• All diseases are fatal. • Examples include **kuru**
Virus	• Non cellular • have living and nonliving characteristics. • 20 to 400 nanometres. • A virus is made up of a protective protein coat. • Viruses are unable to replicate on their own and can only reproduce inside host cells.	• Chickenpox • influenza.
Protozoa	• Single celled eukaryotic. • They have a cell membrane, have no cell walls • have membrane bound nucleus and organelles. • Protozoans can reproduce by binary fission or sexual reproduction. • They range from 1-300 micrometres.	Malaria
Fungi	• Eukaryotic organisms, and possesses a cell wall. • Can be either unicellular, (e.g. yeast) or multicellular (e.g. mushrooms) • Reproduce both asexually and sexually. • Fungi range in size, from macroscopic to microscopic.	• tinea pedis (athlete's foot).
Macro-parasite	• Multicellular eukaryotic • Can be divided into endoparasites, and ectoparasites. • Endoparasites live inside the host's body • Ectoparasites live outside.	Disease caused by **Endoparasites**: taeniasis (tapeworm disease)

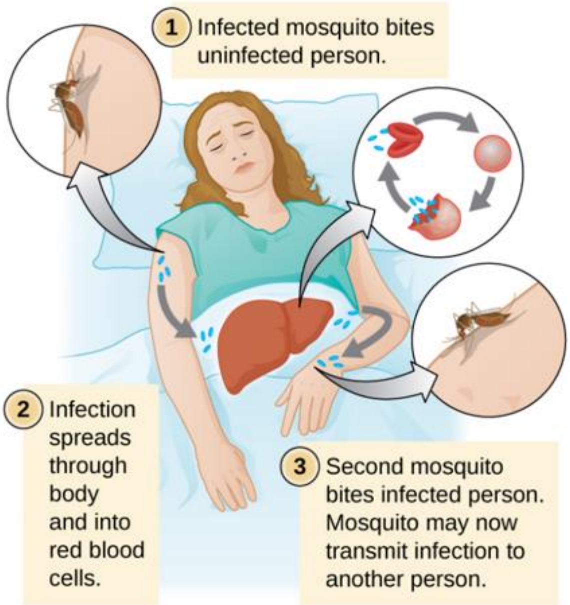


Pathogen adaptations

Using vectors

- Many pathogens can't penetrate host easily, so they have adapted to use vectors, where another organism will transfer them to healthy hosts that they bite/contact
- Most obvious example are mosquitoes, where they transfer the malaria pathogen.





Biochemical adhesion

- **Bacteria develops adhesion proteins** around cell walls that **bind to cell membranes** of the host's cells.
- Then they **secrete toxic chemicals** that **weaken the membrane where they can enter** and infect the host.
- **Viruses** also have a similar adaptation. **They bind to cell membranes** of the host cell and **penetrates it and releases genetic material** into the cell.

Chemical interference

- Some **pathogens can secrete toxins** that **interfere with the hosts signalling system** thus slowing down responses.
- Some **bacterial** pathogens can **break down host's tissue via secretion of chemicals** and **creates entry points** so they can infect cells
- Some **fungi can secrete enzymes** that can **break down cell walls** and membranes of the hosts they're trying to infect.

Response of a named Australian plant against a named pathogen

Plant species

- Eucalyptus tree

Pathogen

- Laetiporus Portentosus

Effect of pathogen on tree

- Can decompose the bark of the tree and cause it to crack.

Tree's response

- Formation of "barrier zones" around infected tissue, protecting healthy tissue from getting infected.
- Barrier zones have a lower pH and moisture content than the regular wood. This limits the amount of nutrition available for the fungus, slowing its growth and development.
- Barrier zones also contain Kino veins that help maintain the structural integrity of the wounded bark.

Innate and adaptive immune system

First line of defence

Skin

- Physical barrier against pathogens (prevents entry)
- Top layer (epidermis) is made of dead skin cells, thus no nutrients + water
- Secretes waxy substance called sebum, another barrier against pathogens

Mucous membrane

- Sticky fluids covering some areas such as tongue and respiratory tract.
- Blocks, and traps pathogens

Cilia

- Tiny hairs that line mucous membranes.
- Pushes mucous outside where it can be coughed up and removed.

Chemical barriers

- Acid in stomach is a big stomach, that destroys most pathogens due to high acidity.

Second line of defence

Inflammation

- Tissue in area of injury releases bradykinin, causes mast cells to release histamines
- Combination of the two causes vasodilation (swelling of capillaries + increased permeability)
- Allows plasma to flow into interstitial space
- Enlarged capillaries + increased blood flow causes the reddening and swelling of the area.
- Increased blood flow → increased temperature → unsuitable for pathogens

Phagocytosis

- Engulfing and digestion of pathogens
- Carried out bu neutrophils and macrophages
- Pathogen is absorbed, lysozymes in the WBC break it down

Cell death

- Cell surrounding pathogens die off and seal off the pathogen-forming a granuloma.
- Protects rest of the body from infection.

Lymph system

- Returns intercellular fluid to circulatory system
- Assists with 3rd line of defence by producing certain types of white cells
- Collects fat absorbed by small intestine

3rd line of defence

Helper T cell triggers both the humoral and cell mediated response. It does not carry out the processes directly, but it delivers a series of chemical signals that initiates the responses.

For those responses to occur:

1) When an antigen binds to antigen receptor of the t cell and is on the surface of an antigen presenting cell, helper t cell multiplies forming clones of activated helper t cells.
2) B cells present antigens to activated T cells, which then activate the B cells.
3) Some activated helper T cells do not release signals but remain dormant for a long time to respond to a future infection of the same pathogen.

Humoral immune response

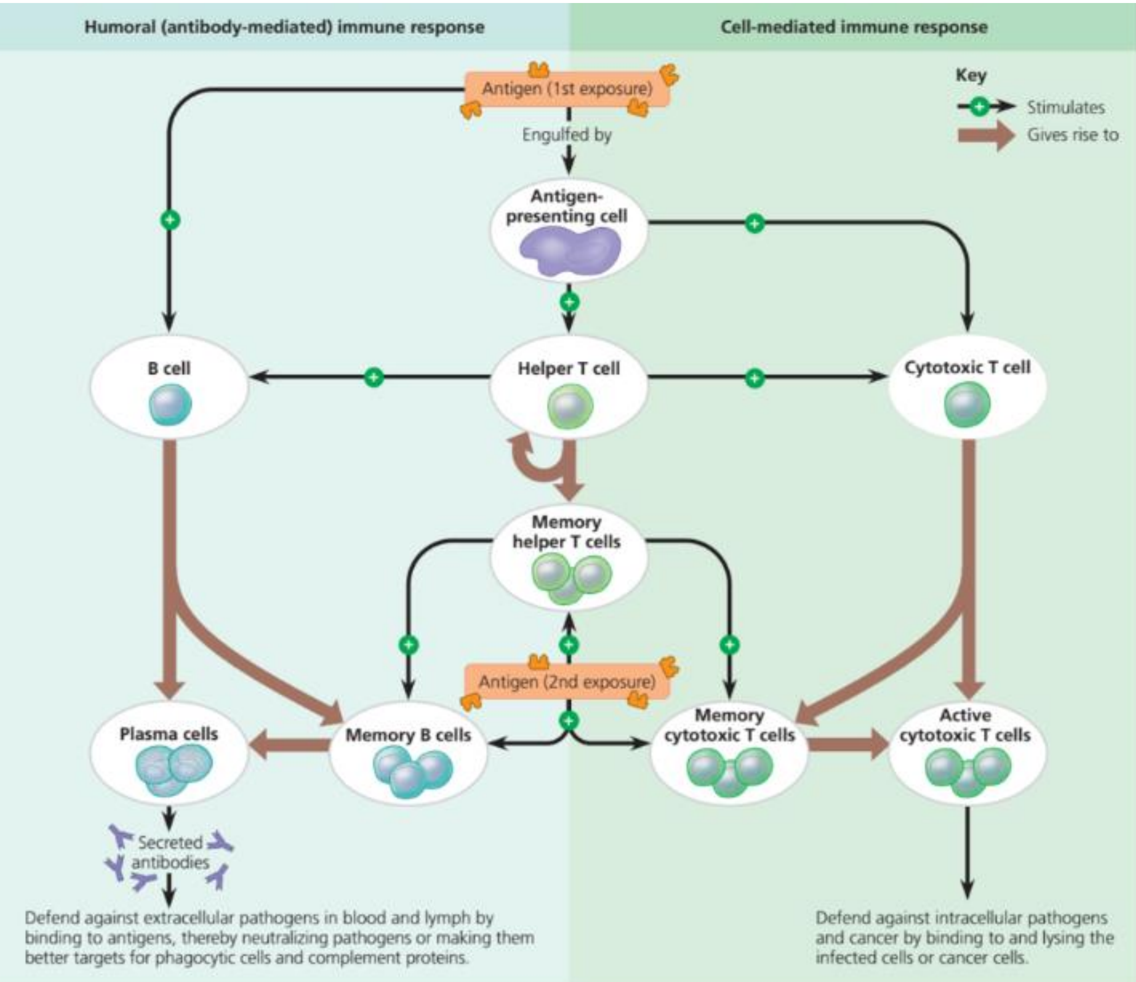
1) B cell is stimulated by cytokines released by activated helper T cell and an antigen, causing it to proliferate and differentiate into memory B cells and plasma cells.
2) Plasma cells produce antibodies which mark pathogens for inactivation

Cell mediated response

1) Cytokines released from helper T ells and interaction with an antigen presenting cell, causing the activation of cytotoxic T cells.
2) Cytotoxic T cell carry target infected cells and destroy them by secreting proteins that triggers apoptosis (cell death).

Memory cells

- B cells, cytotoxic t cells, helper T cells may all appear in memory form, where they lie dormant for many years and only activate upon detection of a reinfection of the same pathogen. Second response is much quicker.



Acquired immunodeficiency virus (AIDS)

Overview of HIV/AIDS

Acquired immune deficiency syndrome or AIDS, is an infectious disease caused by the Human Immunodeficiency virus (HIV). AIDS is the most advanced stage of HIV infection and primarily infects individuals with a weakened immune system and has several specific illnesses. AIDS is a chronic and a life threatening condition. HIV attacks a specific type of cells in the immune system: helper T cell. The helper T cell is vital in the immune responses of the immune system, and by damaging and/or destroying these cells it makes the body much more susceptible to other infections.

Symptoms of AIDS

- Extremely fast weight loss
- Prolonged swelling of the lymph glands in the armpits, groin, or neck
- Pneumonia
- lesions on or under the skin or inside the mouth, nose, or eyelids

Transmission of AIDS

- HIV may be present in any bodily fluid that contains appropriate host cells such as blood, semen, pre-seminal fluid, rectal fluids, vaginal fluids.
- HIV must come into contact with a mucous membrane or be directly injected into the bloodstream for AIDS to be transmitted.
- Most common method of transmission is having unprotected oral, anal or vaginal sex with an infected person. Genital secretions may potentially have HIV and unprotected sex is directly transmitting it to the mucous membranes of the genitalia.
- Another method of transmission is sharing blood products e.g. needles, or unscreened blood. If you share a needle with someone who is infected with HIV, there is a likely chance that their infected blood is on the needle thus injecting the virus, directly into your bloodstream.

Treatment of AIDS

- No cure for AIDS currently.
- Current treatment is effective in only reducing HIV levels in the body and its symptoms and maintaining the health of the immune system.
- Treatment is called antiretroviral therapy (ART) – they take a special combination of HIV medicines every day that mainly prevents the replication of HIV. This doesn't cure AIDS but reduces risk of transmission + increases life expectancy.

Prevention of AIDS.

- Abstinence -No sex with anyone of anonymous nature or unknown history, or high risk individuals e.g. drug abusers or HIV positive people.
- Condoms: covering over the genitals that prevents the exchange of genital fluids.
- Hospital policies: Hospitals follow strict infection prevention methods and control guidelines. All body fluids are treated as infectious and treated as such.

Module 8

Homeostasis

Definition

Ability of certain animals to maintain a stable internal environment regardless of changes to the external environment.

Negative feedback

System works to revert changes in the monitored variable in the system.

Positive feedback

System works to amplify changes in the monitored variable of the system.

STIMULUS	**S**teve
RECEPTOR	**R**ogers
AFFERENT NEURONS	**A**lways
HYPOTHALAMUS	**H**as
EFFERENT NEURONS	**E**veryone
EFFECTOR CELLS	**E**at
RESPONSE	**R**ice

Increase in ambient temperature

1) **Stimulus**
   Increased ambient temperature
2) **Receptor**
   Thermoreceptors in the skin detects the change in external environment
3) **Afferent neurons (PNS)**
   Sends message to control centre about the stimulus via electrical signals
4) **Control centre (CNS)**
   In this case, hypothalamus. Determines the response required in counteracting the change.
5) **Efferent neurons (PNS)**
   Sends message from control centre to effector cells via electrical signals.
6) **Effector cells**
   Sweat glands and blood vessels generate a response to counteract the change
7) **Response**
   Sweat is produced to cool the body via evaporation, which carries heat away from the body. Blood vessels close to surface of the skin vasodilate to allow more blood flow to the surface where heat can be released.

PNS = peripheral nervous system CNS = central nervous system

Increase in blood glucose

1) **Stimulus**
   Rise in blood glucose levels
2) **Receptor**
   Pancreas detects the change in glucose levels
3) **Afferent neurons (PNS)**
   Sends message to control centre about the stimulus via electrical signals
4) **Control centre (CNS)**
   In this case, hypothalamus. Determines the response required in counteracting the change.
5) **Efferent neurons (PNS)**
   Sends message from control centre to effector cells via electrical signals.
6) **Effector cells**
   β cells produce insulin into the circulation
7) **Response**
   Liver and muscle cells take up extra glucose from the blood.
   Glucose is converted to glycogen and stored in the liver. (glycogenesis)

β = bigger glucose = more insulin = increase uptake of glucose + glucose → glycogen (glycogenesis)

α = lower glucose = produce glucagon = converts glycogen to glucose (gluconeogenesis)

Plant adaptations regarding water loss

- **Thick waxy layer**
- **Needle like leaves**
  - Reduced surface area on the leaves thus reduced stomata on each leaf
- **Leaves with hair like fibres**
  - Traps air close to the leaf. As water is lost around the leaf, hairs prevent the water from being blown away.
- **Leaves that can be folded**
- **Leaves that change orientation**
- **Stomata that can be sunken**
- **Reduced number of leaves**
- **Leaf loss during dry season**

Epidemiology

Definition: scientific study of patterns in data to help determine the cause of disease and effectiveness of preventative strategies.

Analysis of data allows the identification of patterns and trends in the incidence, prevalence, and mortality rates of the disease. This info can be used to determine ways in which the overall health of the public can eb improved.

Cohort study

- Involves studying 2 or more groups of people who are free of the disease. These groups differ in their exposure to the potential cause of the disease.
- One of the groups is exposed to the potential cause of the disease and the others are not. They are observed over a long period of time, to compare the resulting incidence of the disease.

Case control

- Case control studies compare people with the disease to people without the disease and look for differences in exposure to the possible causes of the disease.

Intervention studies

Used to test the effectiveness of a treatment or a public health campaign.
Aim is to change the behaviour of the population to reduce the incidence of the disease.

- **Experimental study.**
  - Used to test the effectiveness of a new drug.
  - People suffering from a disease are put into two groups. One group receives the trial drug, and the other receives a placebo.
- **Quasi-experimental**
  - Similar to experimental, but researcher chooses the subjects who receive the drug/placebo.

Treatment of environmental disease: Melanoma

- The treatment for melanoma depends on the stage of the disease and the severity of symptoms. Surgery is the main initial treatment, and involves removing tumour and surrounding skin to ensure to cancer cells remain.
- Additional treatments can include chemotherapy, radiotherapy, targeted therapy and immunotherapy.
- Future directions for further research include development of antibody targeting the CDK4 antigen in order to treat advanced melanoma.

Incidence

- Global incidence has increased steadily, especially in countries with lighter skinned individuals.

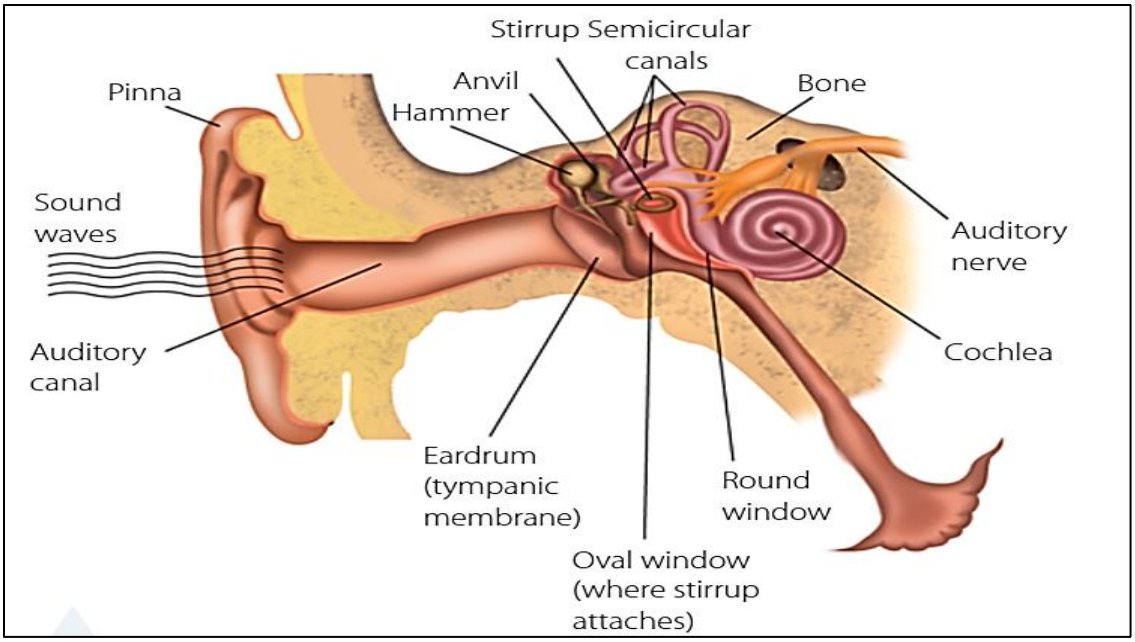
Mortality

- Decreasing mortality rates due to improvements in effectiveness of treatment.

Treatment of nutritional disease: Diabetes

type 2 diabetes is a largely preventable disease often caused by modifiable risk factors.
It can be managed initially through changes in lifestyle, including a healthy diet to manage blood glucose levels and body weight, and regular exercise to help insulin work more effectively.
Future directions for further research include improving gut health.

Structure + Function of ears



Organ	Structure	Function
	Outer ear	
Pinna	Fleshy part of the outer ear	Collect sound waves and channels them into the ear canal.
Auditory canal	Tunnel-like shape	Causes the ear drum to vibrate
Eardrum (tympanic membrane)	Thin skin separating outer and middle ear	Vibrates at the same frequency of the incoming sound waves. Sound waves in air → vibrations in a solid
	Middle ear	
Ear ossicles: Malleus (hammer), Incus (anvil), Stapes (stirrup)	Three bones in the middle ear Stapes is the smallest bone in the body	Transmits and amplifies vibrations from outer ear to inner ear. Tiny bones act as levers and amplify sound by up to 22 times.
Oval window	Membrane separating middle ear and inner ear. Attached to the stirrup.	As stirrup pushes against the oval window, the vibrations are transmitted through to the fluid of the inner ear.
	Inner ear: within the bone of the skull. Extends from the oval window to the cochlea.	
Semicircular canals	three semicircular, interconnected tube	Involved in balance
Cochlea	Spiral tube, separated into three parts by two membranes	Organ of hearing. Converts physical sound waves into electrical impulses.
Organ of Corti	Structure in the middle canal of the cochlea containing the ear receptors in the form of hairs. Hair cells have cilia that are in touch with the tectorial membrane.	Vibrations on the oval window are transferred into waves in the liquid filled cochlea. Receptors (hair cells) are stimulated and fire a nerve impulse to be interpreted by the brain.

Types of hearing loss + Technologies to help

Conductive hearing loss

- Occurs when there's **damage/blockage to the outer/middle ear**.
- This occurs **results in sound not being able to be directed properly** from the ear canal to the eardrums.
- Can be caused by **accumulation of earwax, perforated eardrum**, build up of fluid in the middle ear, abnormal borne growth, etc.
- Common in children and indigenous population.

Technologies for conductive hearing loss

Hearing Aid

- Takes sound waves entering the ear, **increases volume** and directs back into the ear.
- **Cannot improve hearing** when there is damage to the inner ear.
- They're essentially amplifiers, making sounds in environment louder.
- **Relatively cheap** devices, but some people may find them annoying as even **background noises are amplified**

Sensorineural hearing loss

- Occurs when there's **damage or malfunction of hair cells in the cochlea** of the inner ear.
- **Most common type** of permanent hearing loss.
- Causes include **aging** (as hair cells may become more inflexible), **head trauma,** and prolonged exposure to **loud noises.**

Technologies assisting sensorineural hearing loss

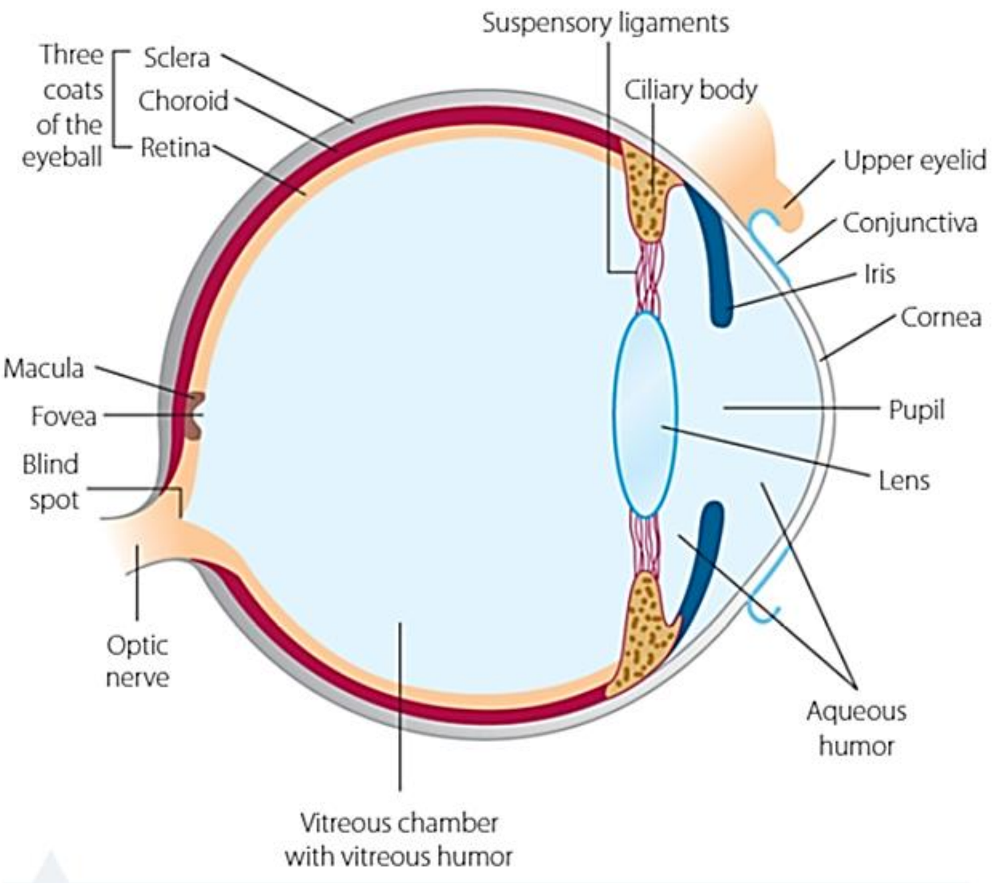
Bone conduction implants

- **Requires surgery** to be installed
- Consists of microphone and processor. Device is able to **transmit vibrations directly to the bone**, thus **bypassing the middle and outer ear** and vibrations arrive directly at the cochlea.
- More useful for permanent hearing loss, where temporary devices may be inconvenient.
- **More expensive procedure** and carries risks with surgery.

Cochlear implants

- Both **external and surgically implanted components**
- **Microphone picks up the sounds** and sends it **to speech processor**.
- Speech processor **converts the sounds to electrical signals** and transmits them **to the magnetically attached receiver**.
- The **receiver transfers the signals to electrodes**, which have been **surgically installed** in the **cochlea**. The electrodes are **able to emulate the generation of electrochemical signals**, **similar to what the hairs** generate, that can be interpreted by the brain.
- There are risks associated with surgery, and **patient must receive training** and get used to the new method of hearing.

The eye



Lens

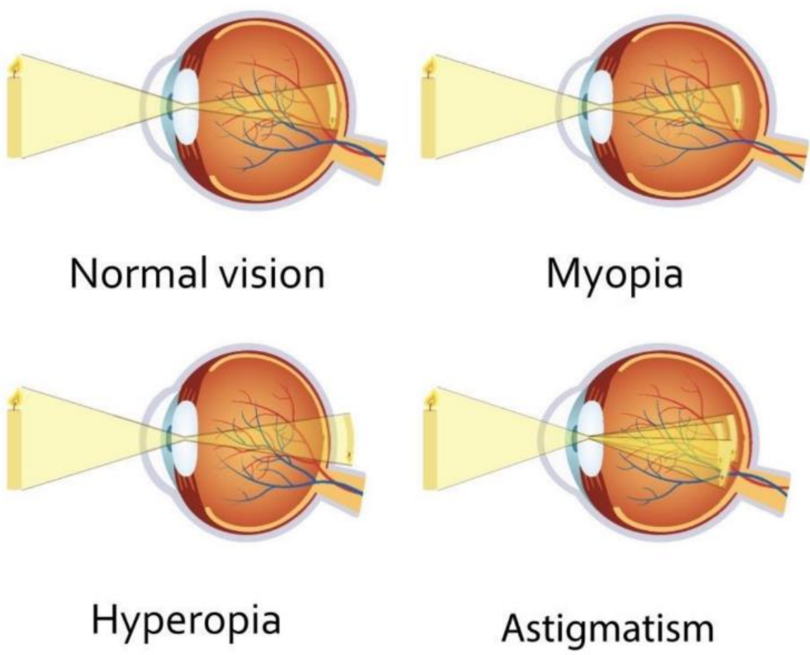
- **Focuses light** from **outside onto the photoreceptors**
- Ability for the eye to focus on near and far objects is due to contractions of the ciliary muscles in the ciliary body.
  - When **ciliary muscles contract**, suspensory **ligaments lose hold of the lens**, and lens **becomes more rounded**. Refraction of light is increased, and **near objects** would be in **focus**.
  - When **ciliary muscles expand**, suspensory **ligaments are taut**, and **lens is flattened**. Refraction is at minimum, and **far objects** would be **focused**.
- Accommodation: ability for the eye to change focus to see at varying distances.

Photoreceptors in the Retina

- Light strikes the retina, passes through highly transparent neurons, and reaches two types of photoreceptors: rods and cones.
- **Rods**
  - Photoreceptor that **detects light** but does **not distinguish colours**. Responsible for **night vision**.
- **Cones**
  - **Provides colour vision**. Provide very little in night vision. Three types: **L (long λ) cones**, **M (medium λ) cones**, **S (short λ) cones**. **Most cones are concentrated** in the region of the retina called the **fovea**.
  - When light hits the eye, more than one type of cone will be stimulated. Retina and the brain process the mixture so that different hues and intensities can be stimulated.
- Photoreceptors are distributed unevenly throughout the retina. There is one region of the retina called the optic disc, where the nerve fibres leave the eye and converge to form the optic nerve, has no photoreceptors, creating a blind spot.

Refractive Visual disorders

Many visual disorders are the result of incorrect focusing of light. In normal vision, lens adjusts its shape to refract light so that the focal point is focused at region around the fovea, where most cones are located.



Myopia (Short-sightedness)

- Can't focus on far away objects
- Light from distant objects are refracted more than necessary, so focal point is in front of the retina.
- Can be caused when ciliary muscles fail to expand properly, thus affecting the refracting of the lens and causing farther objects to be blurry.
- Can also be caused by individual just having longer eyeball

Hyperopia (Long-sightedness)

- Can't focus on nearby objects
- Lens can't assume the rounded shape required when viewing nearby objects. Causes images to be focused at a point behind the retina.
- Can be caused by ciliary muscles failing to contract when viewing nearby objects. Results in the lens being too flat, and light is not being refracted enough.
- Can also be caused by individual having shorter eyeball, (shorter axial length).

Astigmatism

- Blurred vision at all distances
- Occurs when either the front cornea or the lens have mismatched curves.
- Surface of the eyeball is egg shaped

Technologies for refractive errors

Glasses/contact lenses.

Light can be **artificially refracted by wearing a lens** which can diverge the light rays just enough so that it focuses on the retina**.**

- **Correcting myopia**
  - Nature of the problem is that light **is focused in front of the retina.**
  - Thus **correction requires a diverging lens**, i.e. a **concave** lens.
- Correcting **hyperopia**
  - Light is **focused behind the retina**
  - Corrected by a **convex** lens.

Both methods refract light before it enters the eye.

Laser surgery

- Used to **change the shape of the cornea permanently** so that it converges or diverges the light to correct any refractive condition.
- **A flap of the cornea is cut** and lifted out. **A laser beam reshapes the rest of the cornea**. The flap is then replaced.
- A more expensive solution than using lenses but is more permanent and convenient than having to wear glasses or contacts.

Other Visual Disorders

Cataracts

Lens is mostly made of water and crystalline proteins, which are arranged to allow light to pass through. Sometimes the proteins clump together in areas of the lens causing vision problems known as cataracts.

- Cataracts is the **general clouding of the lens over a prolonged period of time**.
- Treatment involves **replacing the cloudy lens** with an **artificial intraocular lens (IOL**).
- Phacoemulsification: most common technique for implanting the IOL.
  - Small incision is made where the cornea meets the sclera.
  - Lens is then suctioned out and replaced with IOL. Incision is very small and is self-healing so no stitches required.

Colour blindness

Full colour vision depends on having all three types of cone cells being present and functioning properly. Defects in one or more of these cone cells will affect colour sensation, resulting in colour blindness.

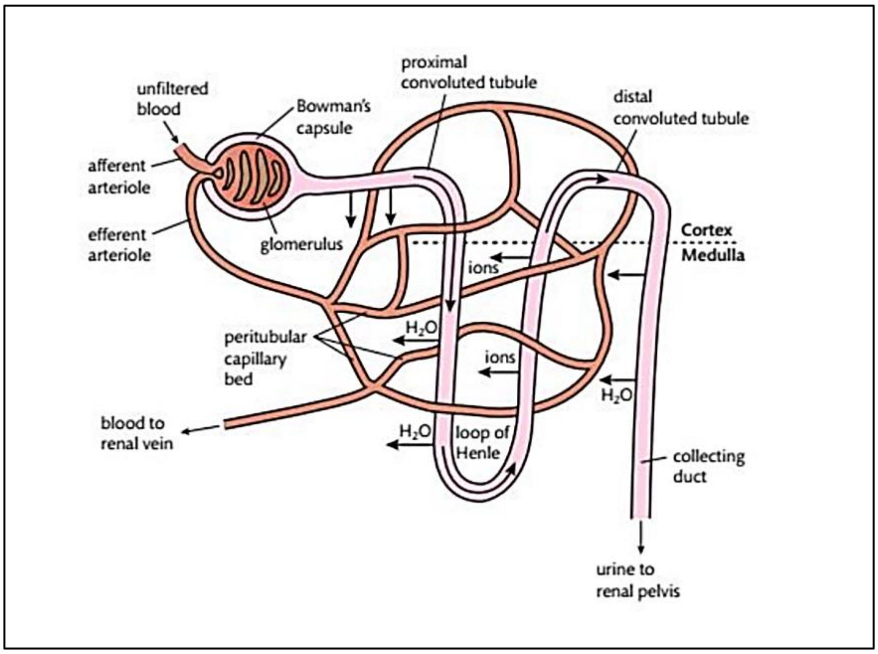
- Occurs when one or more of the cones are missing or defective.
- Thus they're unable to manufacture necessary signals to the brain.
- Causes are genetic. Most common type of colour blindness: red/green blindness is due to a defective x-linked allele. More common in males.

The kidney

Functions of the kidney

- Filters blood by removing waste from blood flowing through.
- Vital in osmoregulation: control of salt and water balance in the body to maintain a constant internal environment in the body. Achieved by regulation of volume of urine produced in the kidneys.
- May also regulate blood pH by secreting excess H ions into waste filtrate.

Nephron



Glomerulus (surrounded by bowman's capsule) → proximal convoluted tubule→loop of Henle→distal convoluted tubule→ collecting duct

Chronic kidney disease

- Most common form of kidney disease. Long term condition and doesn't improve over time.
- Main causes include high blood pressure and diabetes
- High blood pressure
  - Dangerous for kidneys as it increases pressure on the glomerulus of the nephron. Over time, increased pressure can damage the vessels deteriorating kidney function.
- Diabetes
  - High levels of sugar in the blood can damage the blood vessels in kidneys over time.
- Kidney failure as a result of chronic kidney disease is the final stage and occurs when waste accumulates in the body as a result of long term kidney malfunction.

Renal dialysis

Artificial method of filtering the blood. Used when someone's kidneys have failed or are close to failing as a result of chronic kidney disease. **Two types** of dialysis: **haemodialysis** and **peritoneal dialysis.**

Haemodialysis

- Blood is pumped through a special machine that filters out the waste products and fluid.
- Is carried out in hospital or in professional environment.
- Can be inconvenient as patient must stay still for up to five hours.

Peritoneal dialysis

- Peritoneum (membrane lining the abdominal wall) stands in for the kidney.
- Tube is implanted and fills the abdomen with dialysate. Waste flows from peritoneum to the dialysate.
- Dialysate is then drained out of the abdomen.
- Can be done independently at home.

Evaluation of a technology (Cataract surgery)

Cataract treatment via implantation of the IOL is extremely effective.

- Surgery is relatively simple and quick, allowing patients to return to their regular lives quickly.
- It is a surgical procedure, and with all surgeries, there is some risk involved. However the IOL replacement is considered one of the safest types of surgery.
- it eliminates the cataract problem, but can be quite expensive and people in developing countries without proper healthcare may not be able to afford it.